

# Benjamin Kim

[benkim0634@gmail.com](mailto:benkim0634@gmail.com) | [linkedin.com/in/benjamin-kim1](https://linkedin.com/in/benjamin-kim1)

## EDUCATION

**University of California San Diego**

*Bachelor of Science in Computer Science*

*NCAA Division 1 Fencing Team*

GPA: 3.85

*(Expected) May 2027*

*September 2024 - Present*

**Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Computer Organization, Systems Programming, Discrete Mathematics, Linear Algebra, Calculus, Statistical Methods

## EXPERIENCE

**Association for Computing Machinery**

Sept 2025 - December 2025

*Full-Stack Developer | React, Node.js, Express, Typescript, Discord + Stripe API University of California, San Diego*

- Developed a campus marketplace delivery platform using **React** and **Express**, connecting students with deliverers via a web app and Discord bot integration serving **20+** users
- Built a secure payment processing system with **Node.js** and **TypeScript**, integrating the **Stripe API** and implementing pending transaction logic to prevent fraud and ensure deliverer compensation.
- Collaborated with cross-functional team of **5** to define project scope and deliver features on schedule

## PROJECTS

**UCSD Fencing Attendance App** | *Next.js, TypeScript, PostgreSQL, Tailwind CSS* August 2025 - October 2025

- Developed full-stack attendance platform with **Supabase/PostgreSQL** backend, implementing role-based access control and **NextAuth** authentication to manage **40+** active fencers across practice and gym training sessions
- Built **REST APIs** with composite unique keys to enable accurate tracking of practice and gym sessions
- Designed interactive weekly and monthly dashboards with recurring and custom scheduling, using client-side caching and live updates to deliver real-time data for easy user interaction

**AI-Powered Text Simplifier | Chrome Extension**

October 2025 - November 2025

- Developed a **full-stack Chrome extension** that simplifies and summarizes highlighted webpage text using a Flask backend integrated with **OpenAI's GPT-3.5 API**, improving user reading comprehension
- Designed a frontend UI with **HTML/CSS** and implemented communication with the backend using **Javascript**
- Implemented secure **Flask** backend with input validation, error handling, and protected API key management

**Multi-user Chat Application** | *Python, Tkinter, SQLite*

October 2023 - December 2023

- Developed concurrent **TCP** chat server using Python threading and sockets to support **10+** simultaneous client connections with message broadcasting and private messaging capabilities
- Designed normalized SQL schema with indexed message history and persistent chat logs across user sessions
- Created responsive **Tkinter** GUI with auto-scrolling message feed and asynchronous message handling for smooth multi-user chat experience

**Computer Vision & ML Projects** | *Python, TensorFlow, OpenCV, Keras*

November 2023 - January 2024

- Built image classifiers for MNIST and Fashion-MNIST datasets (**70K** images), training neural networks that achieved **97%** test accuracy on handwritten digit recognition
- Developed image preprocessing pipeline using **OpenCV** to detect contours, extract regions of interest, and make predictions on external images with bounding boxes for visualization

## TECHNICAL SKILLS

**Languages:** Java, Python, JavaScript, HTML, CSS, SQL, MATLAB

**Libraries & Frameworks:** React.js, Node.js, Next.js, Flask, TensorFlow, Keras, PyTorch, NumPy, Pandas, OpenCV, Matplotlib

**Developer Tools:** Git, Supabase, PostgreSQL, Tkinter